



**114** Thin Is In

# Digital Business

How Technology Can Help Your Organisation

## Tech Careers

# Chip Your Way Through!

**VLSI design is a lucrative career prospect. It's also not for the faint hearted**

**Samir Makwana**

**C**hip designing in India began when Texas Instruments started its operations in 1985 in Bangalore. Many multinational companies have since followed, and there are quite a few Indian companies that have also begun VLSI chip designing. We now have home-grown intellectual property, including digital, analogue, mixed-signal, and RF technologies being developed to meet the demands of electronics manufacturing companies from around the globe.

Outsourcing chip design from the US was done for the advantages of lower overheads and labour costs. Also, time-to-market integrated device manufacturers and ASIC (Application-Specific Integrated Circuit) vendors utilise the expert services of integrated design houses to design their chips.

Atul Arora, President—Commercial Relations, ARM Embedded Technologies Pvt Ltd, says, "The advantage ARM has by having

operations in India is the ability to tap into the talent pool of Indian engineers. That gives us the long-term investment ability to nurture and train engineers in ARM-based architecture for the development of innovative products."

About 120 semiconductor players engaged in chip designing are contributing to the Indian industry. According to a report by The ISA and Frost & Sullivan, the Indian chip design industry is expected to derive an output of more than Rs 150 crore in the financial year 2005-06. So what does it take to get into this field? What options are available? Let's have a quick look.

## The Basic Requirements

The chip design industry is still in its early stages, and is set to grow. An important aspect is that in this field, demand is higher than supply—a rarity indeed. "The first and foremost requisite (to enter this field) is to have an early understanding of the industry, rather than just deciding to become a

Illustration: Chaitanya Surpur

